INCH-POUND

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MILITARY SPECIFICATION

SHEARING, NIBBLING, AND FORMING MACHINES; UNIVERSAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of universal shearing, nibbling, and forming machines designed to perform metal working operations of shearing, nibbling, piercing, punching, slotting, beading, louvering, dishing, flanging, and bending.

1.2 Classification. The machines shall be of the following types. The sizes shall be as listed in tables I and II. The type and size to be furnished shall be as specified (see 6.2.1).

Type I - Straight and circle shearing, nibbling, and forming, accomplished by utilizing straight and circle guiding attachments to guide the workpiece through metal working operations.

TYPE II - Straight, circle, and contour shearing, nibbling, and forming, accomplished by utilizing either a template duplicating attachment, or a coordinate guide system to guide the workpiece through metal working operations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3445

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2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government document. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise

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specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.1 - Mechanical Power Presses-Safety Requirements for the Construction, Care, and Use.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Scales Departments, 1430 Broadway, New York, NY 10018-3363.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101-4269.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - MOTORS and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturers' current production models, designed to perform sheet metal shearing, nibbling, slotting, beading, louvering, dishing, flanging, and bending operations required herein. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Design, construction, and system of measurement. Unless otherwise specified, either the inch/pound system or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in inch/pound units. These dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the inch/pound or the SI system of measurement shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as

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specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 6.2.1), measuring and indicating devices shall be graduated to read in both the inch/pound and the SI system of measurements.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the intended use, performance, and design life of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station.

3.2.5 Safety and health requirements. All parts, components, mechanism, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all requirements of 29 CFR 1910 and ANSI B11.1. If a conflict arises between 29 CFR 1910 and B11.1, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified in detail (see 6.2.1).

3.2.5.1 Safety guarding. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance, and repair of the parts. The safety devices shall not interfere with the operation of the machine. If specific point-of-operation guarding dependent on workpiece size and configuration is required, the procuring activity shall specifiy the exact configuration of the guard required (see 6.2.1)

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operation, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation .

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. When specified (see 6.2.1), an automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir

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shall be by an oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each system shall include a cleanable or replaceable filter.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standards" section referenced therein.

3.3.7 Dials. The diameter of the dials shall be such that graduations may be read from the operator's normal position. Dials shall be permanently and legibly engraved or etched on a nonglare background. Graduations indicating stock removal shall be in increments of not more than 1/32 inch.

3.3.8 Gears. All gears shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the inch/pound system or ISO 54 for the SI system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened steel. Working surface hardness shall be not less than Rockwell C-48.

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3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The machine shall consist of not less than a frame, machine drive, machining head, tool holders, cutting attachments, electrical system and other components required to meet the requirements specified herein.

3.4.1 Frame. The frame shall either enclose or support the machining head and upper tool holder, machine drive, lower tool holder, and all other components specified herein. The frame shall have the strength and rigidity necessary to assure the accuracies specified herein and shall be constructed to withstand stresses produced when operating the machine at its maximum capacity without permanent deformation or failure. The frame shall be of "C" style configuration, stress relieved, with the overhang column and base forming a workpiece opening or throat. The depth of the throat shall be as specified for the machine sizes listed in table II. The machine shall be capable of performing the machining operations as listed in table I. The frame enclosure shall have covered access openings of the size required to facilitate servicing, maintenance, and adjustments of all components contained therein. The frame shall be a floor mounting unit, arranged to support the machining tool at a working height of not less than 36 inches, and not more than 45 inches from the mounting surface. The base of the frame shall have means to permit leveling and fastening the machine to the floor or foundation. When specified (see 6.2.1), vibration dampening leveling pads shall be provided. An eye or an attachment point shall be provided at or near its balance point on the frame for lifting the machine. The eye or lift point shall be capable of supporting twice the weight of the machine, and its location shall not cause distortion to the frame when the machine is suspended.

3.4.1.1 Guide rails. The throat of the "C" frame shall be provided with upper and lower guide rails mounted in or along both sides of the throat opening. The guide rails shall provide parallel surfaces along the usable depth of the throat for mounting the work guiding attachments. These attachments shall mount on a saddle or other device which slides on bearing surface ways of the guide rails. The bearing surface of the ways shall have a finish of not more than 63 micro-inches, and the ways shall be flat within a tolerance of + 0.003 inch per linear foot. The rails shall provide gripping surfaces for locking the saddle and securing adjustments for the workpiece attachments. A scale, graduated in increments of not more than 1/32 inch, shall be permanently marked on the guide rails to indicate lateral adjustment of the attachments.

3.4.2 Machine drive. Unless otherwise specified, the machining action of the machine shall be accomplished by either a mechanical type drive or a hydraulic type drive. When only one type drive is acceptable the particular drive shall be as specified (see 6.2.1). The drive shall be designed to assure that sufficient energy is developed for accomplishing the performance requirements specified herein.

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TABLE I. Sizes and performance requirements.

Metalworking operations	Workpiece thickness (inches)								
	Type I					Type II			
	Size 7	Size 8	Size 9	Size 10	Size 11	Size 6	Size 7	Size 8	Size 9
Shearing, edge	.149	.149	.224	.250	.281	.281	.281	.312	.360
Shearing, center (without starting holes)	.119	.119	.179	.188	.188	.188	.188	.250	.312
Nibbling	.075	.075	.119	.156	.179	.156	.188	.188	.200
Slotting	.075	.075	.119	.125	.149	.156	.156	.188	.188
Beading	.104	.104	.125	.135	.135	.135	.135	.156	.188
Louvering	.075	.075	.119	.125	.119	.125	.125	.156	.125
Dishing	.075	.119	.119	.125	.119	.125	.125	.125	.250
Flanging	.075	.075	.075	.075	.105	.105	.105	.119	.119
Bending (up to 90 degrees)	.075	.075	.105	.105	.105	.105	.105	.119	.188

NOTES: The machine shall be capable of performing the metalworking operations in workpieces of mild steel having "not less than" 50,000 PSI tensile strength.

Requirements are "not less than" those shown.

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TABLE I. Sizes and capacity.

Item	Type I					Type II			
	Size 7	Size 8	Size 9	Size 10	Size 11	Size 6	Size 7	Size 8	Size 9
Throat depth of frame, inches, not less than	28	39	41	48	48	41	48	48	60
Circle cutting attachment: Circle diameters, inches; range									
Small - not more than	3-1/4	3-1/4	3-1/8	3-1/8	4-5/16	-	-	-	-
Large - not less than	28	39	40	45	45	-	-	-	-
Template duplicating attachment:									
1/ Duplicating range, Y x X axes, inches not less than	-	-	-	-	-	29x48	36x48	36x78	48x48
2/ Circle cutting capacity, diameters, inches; range									
Small - not more than	-	-	-	-	-	3-3/16	4-5/16	4-5/16	4-5/16
Large - not less than	-	-	-	-	-	29	36	36	48
Machining tool speed, Strokes per minutes; range									
Infinitely variable or Progressively stepped Low - not more than	1700	1700	400	400	350	400	350	350	250
High - not less than	3400	3400	2800	2800	1400	1400	1400	1400	1400
Motor HP, not less than	3/4	3/4	2.0	2.5	2.9	2.5	2.9	2.9	2.9

1/ Duplicating range without repositioning the workpieces.
2/ Circle cutting by nibbling with duplicating attachment.

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3.4.2.1 Mechanical type drive. The mechanical type drive system when furnished shall transfer power from the electric motor to the machining head. The drive shall convert rotary motion to reciprocating motion. The drive shall consist of an electric motor conforming to 3.4.8.2, and a belt drive arrangement powering a driven eccentric system. The belt drive arrangement shall be of the multiple V-belt, timing belt, or poly-belt type, and shall be capable of transmitting full motor torque without belt slippage. Rotating members shall be supported on bearings of a size, type, and precision class selected for the specific application to ensure maximum service life and accuracy. The drives shall be enclosed in and supported by the machine frame. The drive shall provide the upper machining tool with the number of reciprocating strokes required for the high and low speed requirements of table II. The machine shall provide means to adjust stroke rate without stopping the machine.

3.4.2.2 Hydraulic type drive. The hydraulic type drive system when furnished shall be complete, including all pumps, valves, piping, ram cylinder unit, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to hydraulic components. The ram cylinder unit shall be provided with overload protection by a pressure relief mechanism on the ram. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have a means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable. The drive shall provide the upper machining tool with the number of reciprocating strokes required for the high and low speed requirements of table II. A means shall be provided to adjust stroke rate without stopping the machine. The system shall conform to the requirements of JIC H-1 Hydraulic Standards.

3.4.3 Machining head. The machining head shall include a separate upper and lower assembly, supported respectively by the overhang and base of the frame. The upper assembly of the machining head shall be provided with components to convert the rotary motion of the drive to the reciprocating motion required for accomplishing the machining operations specified herein. The machining head shall be provided with means for adjusting the length of the reciprocating stroke. The upper assembly of the machining head shall be provided with an upper tool holder, and the lower assembly shall be provided with a lower tool holder. The reciprocating motion of the machining head shall be controlled by either a hand or foot operated drive control switch located in a position for actuation at the operator's station. Selection of the speed range shall be by a hand lever or other means.

3.4.3.1 Tool holders. The tool holders shall be designed to accommodate the tools required for performing the machine operations specified in table I. The tool holders shall be of the quick change type, and shall be designed to hold the machining tool in position to prevent movement other than that required for the machining operations. The tool holders shall have means for adjusting the height between the tool points, and for maintaining alignment after it has been set between the machining tools. All tools shall have a common shank dimension of either square or round shape. Tools that require a collet or sleeve shall be supplied with the cutting tools. Collets on sleeves shall have a flat surface for tool orientation. Collets or sleeves shall locate the tool securely for the machining operations.

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3.4.4 Straight cutting attachment. The type I machine shall be furnished with a straight cutting attachment. The attachment shall be provided with a straight edge for guiding the workpiece through the cutter tools without binding to obtain precision straight shearing as specified herein. The attachment shall be provided with machined bearing surfaces for slide adjustment of the straight edge along the guide rails of the frame. The bearing surfaces shall be machined to a finish of not more than 63 micro-inches, and shall be flat to within a tolerance of ± 0.003 inch per linear foot. Means shall be provided for locking the straight edge in position. The straight cutting attachment shall guide the workpieces for cutting strip width sizes from one inch through any size up to a measurement equal to 3/4 of the throat depth of the frame.

3.4.5 Circle cutting attachment. The type I and type II machines shall be furnished with a circle cutting attachment. The attachment shall be provided with clamping shoes for holding and pivot guiding the workpieces to produce circular cuts. The attachment shall be provided with machined bearing surfaces for slide adjustment of the clamping shoes along the guide rails of the frame. The bearing surfaces shall be machined to a finish of not more than 63 micro-inches or better and flat to within a tolerance of ± 0.003 inch per linear foot. Means shall be provided for locking the clamping shoes in position.

3.4.6 Template duplicating attachment. Unless otherwise specified, (see 6.2.1), the type II machine shall be furnished with a template duplicating attachment for straight shearing, and straight and contour nibbling. The attachment shall be a pantograph style arrangement with a manually controlled stylus for tracing the contours of flat templates. Positioning of the stylus by the operator shall simultaneously position the workpiece and guide the workpiece past the cutting tools duplicating the template. The workpiece carriage shall be provided with replaceable machined bearing surfaces for slide adjustment along the replaceable guide rails of the frame. The bearing surfaces shall be machined to a finish of not more than 63 micro-inches, and shall be flat to within a tolerance of ± 0.003 inch per linear foot. Means shall be provided for locking the workpiece carriage in position. The attachment shall be provided with a template holder for securing the template in the duplicating position. The attachment shall be furnished with scales, calibrated in increments of not more than 1/32 inch, to provide coordinate positioning and nibbling directly from part drawings.

3.4.6.1 Coordinate guide systems. When specified (see 6.2.1), the type II machine shall be furnished with a coordinate guide system, with workpiece carriage characteristics as specified in 3.4.6.

3.4.7 Digital readout (DRO) system. When specified (see 6.2.1), a DRO system shall be provided on the Y and X axes of travel of the duplicating attachment. The Y and X axes of travel of DRO system shall be not less than specified in table I. The DRO system shall consist of a position feedback device, an electronic display for each axis specified and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.001 inch. The system accuracy shall be 0.001 inch per foot and the repeatability shall be 0.001 inch with a

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nonaccumulative tolerance of 0.005 inch over the full duplicating range of travel. Special requirements for the DRO system components shall be specified (see 6.2.1). The DRO shall indicate axis location in both the inch/pound and SI units. The DRO system shall be provided with electronic and manual calibration to the machine on which it is installed to eliminate the effects of distortion error.

3.4.7.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and inadvertent damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination of oil, dust, chips, coolants, and airborne particles. Position feedback devices shall be easily and quickly interchangeable between like machines in any multiple machine order.

3.4.7.2 Electronic display console. The electronic display console for the DRO system shall contain a universal digital display with individual digits not less than 0.56 inch high with an effective digit height of not less than 0.43 inch. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operations of the machine. The cable shall be securely fastened to the structure of the machine in accordance with the requirements of the contract.

3.4.8. Electrical system. The electrical system shall conform to NFPA 79. Each machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.8.1 Primary input voltage. Unless otherwise specified (see 6.2.1), the machine shall be wired to draw all of its electrical power from a single 230/460 volt, 3-phase 60 Hz circuit. Unless otherwise specified (see 6.2.1), the machine shall be initially wired to operate from a 460 volt source.

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3.4.8.2 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 50/60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of NEMA MG-1.

3.4.8.3 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.8.4 Worklight. When specified (see 6.2.1), an incandescent worklight shall be provided. The fixture shall be permanently mounted to the machine with air vented protective reflector shield, on-off switch, and shall be adjustable to reflect maximum light on the work area. The lamp shall be no less than 75 watt.

3.4.9 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table II for the machine size specified.

3.6 Performance. The machine shall be capable of performing sheet metal shearing, nibbling, slotting, beading, louvering, dishing, flanging, and bending operations in the workpiece thicknesses specified in table I for the type and size machine required. The machine shall be capable of performing the metal working operations in workpieces of mild steel having not less than 50,000 PSI tensile strength without wrinkling or deforming the workpiece.

3.6.1 Type I machine. The type I machine shall be capable of performing straight shear cuts to within $\pm$ 0.003 inch per linear foot of cut, parallel with a squared workpiece edge, when guided by the straight edge of the straight cutting attachment. The machine shall be capable of performing circular shear cuts of any size ranging from the smallest to the largest diameters specified for the circle cutting attachment in table II.

3.6.2 Type II machine. The type II machine shall be capable of performing straight shear and straight and contour nibbling operations to duplicate flat templates (utilizing the template duplicating attachment or coordinated guide system). The duplicated workpieces shall not deviate more than $\pm$ 0.005 inch per linear foot of cut from guided edges of the duplicated template. The duplicating attachment shall be provided with a duplicating range and a circle cutting capacity to comply with table II.

3.6.3 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

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3.7 Standard equipment. Unless otherwise specified (see 6.2.1), the following tooling shall be furnished with each machine. Tooling shall be compatible to the size machine specified in tables I and II.

a. One set of tools or wrenches normally furnished with the machine.

b. One set of slot cutting tools (straight and circular cutting).

c. One set of nibbling tools (punches, dies, and pressure foot).

d. One set of circle cutting tools.

e. One set of shearing tools.

f. One center tool holder attachment.

g. One tool grinding gage.

h. One each (round and square shape) punch and die set - one inch size.

3.8 Optional equipment. Optional tooling specified in the following subparagraphs shall be furnished with each machine as specified (see 6.2.1). Optional tooling shall be compatible to the size machine specified in tables I and II.

a. One additional set of nibbling tools (punches, dies, and pressure foot).

b. One additional set of shearing tools.

c. One each (round and square shape) punch and die set - one inch size.

d. One set of beading tools, 1/4 inch size.

e. One set of louvering tools, 1/4 inch height.

f. One set of dishing tools (4-14 inch).

g. One set of flanging tools, 3/8 inch flange.

h. One set of 90 degree bending tools.

i. Outside circle guide attachment. Radius of workpiece shall be equal to throat depth).

j. Circle nibbling attachment with graduated angle dial.

k. Air mist tool lubricating system.

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

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3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication plate. A corrosion resistant lubrication plate shall be securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A corrosion resistant nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the machine and accessories shall be commensurate with all requirements specified herein and shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification

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shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of 41 to 104 degrees fahrenheit and 20 to 95 percent relative humidity.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, machining head, machine drive, and attachments shall be verified during the trail period.

4.7.2 Performance tests. The machine's performance shall be tested by operating the machine to perform sheet metal shearing, nibbling, slotting, beading, louvering, dishing, flanging, and bending operations in the maximum workpiece thicknesses to determine conformance with the requirements of 3.6 and table I. The machine shall perform the test operations without wrinkling or deforming the workpiece.

4.7.3 Type I machine. Type I machine shall also be tested by performing straight shear cuts utilizing the straight shearing attachment, and circular shear cuts utilizing the circle cutting attachment to verify the requirements of 3.6.1.

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4.7.4 Type II machine. Type II machine shall also be tested by performing straight shearing and straight and circular nibbling by utilizing the template duplicating attachment or coordinate guide system to verify the requirements of 3.6.2.

4.7.5 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.9 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine conformance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended to incorporate simple design, rapid tool change, and easy tool adjustment for use in prototype or short run sheet metal shearing, nibbling, and forming operations. Different operations may be performed on the same machine without expensive tooling and movement of job to other machines.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2 and tables I and II).

c. When first article is required for inspection and approval (see 3.1 and 4.4).

d. If machine is required to be configured in a specific measurement system (inch/pound or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (inch/pound or SI), state required system (see 3.2.1.1).

f. Dual calibrations (inch/pound and SI), if required (see 3.2.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

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h. Specify configuration of additional point-of-operation guarding, if required (3.2.5.1).

i. Automatic lubrication, if required (see 3.2.9).

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

l. Vibration dampening leveling pads, if required (see 3.4.1)

m. If machine drive is to be a specific type, state required type (see 3.4.2).

n. Template duplicating attachment on type II, if different (see 3.4.6).

o. Coordinate guide system on type II, if required (see 3.4.6.1).

p. Digital readout system, if required; 3.4.7).

q. Specify, and fully describe any special requirement (see 3.4.7).

r. Primary input voltage, if different (see 3.4.8.1).

s. Initial wiring, if different (see 3.4.8.1)

t. Motors, enclosure, if different (see 3.4.8.2).

u. Worklight, if required (see 3.4.8.4).

v. Hour meter, if different (see 3.4.9).

w. Size, and capacity, if different (see 3.5).

x. Additional performance features, if required (see 3.6.3).

y. Standard equipment, if different (see 3.7).

z. Specify and fully describe optional equipment, if required (see 3.8).

aa. Specify applicable repair parts, if required (see 3.9).

bb. Inspection conditions, if different (see 4.3).

cc. First article inspection, if different (see 4.4).

dd. Quality conformance inspection, if different (see 4.5).

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ee. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports are to be specified on the DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First Article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. Specific instructions regarding arrangements for examination, test, and approval of the first article are given in the contract.

6.4 Cross-reference of classification changes. The machine classification changes from the previous issue of this specification reflects changes in all sizes, capacity and performance requirements as follows:

MIL-S-80060C	MIL-S-80060D
TYPE I	TYPE I
Size 2	Deleted
Size 3	Deleted
Size 4	Deleted
Size 5	Deleted
Size 6	Deleted
	Size 7 (New size)
	Size 8 (New size)
	Size 9 (New size)
	Size 10 (New size)
	Size 11 (New size)
TYPE II	TYPE II
Size 1	Deleted
Size 2	Deleted
Size 3	Deleted
Size 4	Deleted
Size 5	Deleted
	Size 6 (New size)
	Size 7 (New size)
	Size 8 (New size)
	Size 9 (New size)

6.5 Subject term (key word) listing.

Forming
Nibbling
Shearing

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Safety health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity is advised to analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.8 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.9 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activity:
Air Force - 84

User activity:
Navy - MC

Preparing activity:
DLA-IP

(Project 3445-0112)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-80060D	Shearing, Nibbling, and Forming Machines; Universal

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

PREVIOUS EDITION IS OBSOLETE.

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